

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	xxxxxx
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	The primary objective is to develop an AI-based system that predicts an individual's overall well-being using Machine Learning, enabling users to better understand their health status and receive personalized recommendations for improvement.
Scope	The project includes data collection, preprocessing, feature selection, model training using Random Forest, well-being score prediction, visualization of results, and deployment through a Flask-based web application for real-time user interaction.

Problem Statement	
Description	Many individuals struggle to evaluate their overall well-being as existing solutions focus on only one aspect of health, such as fitness or mental health. Manual assessments are time-consuming and do not provide a comprehensive and data-driven evaluation.
Impact	Solving this problem will help users monitor their overall well-being accurately, make informed lifestyle decisions, improve quality of life, promote preventive healthcare, and assist healthcare professionals with faster and more reliable assessments.

Proposed Solution	
Approach	Develop a Machine Learning model using Random Forest to analyze health, lifestyle, and socio-economic factors. The system predicts an individual's well-being score and category, and provides personalized recommendations through an interactive web application with visualizations.
Key Features	- User-friendly web interface using Flask and Bootstrap. - Secure user input collection and validation. - Data preprocessing, feature selection, and model training. - Well-being prediction using Random Forest algorithm. - Interactive dashboard with charts and visual insights. - Personalized recommendations based on prediction result. - Model saving using Joblib and real-time prediction.

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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU for model training and prediction	Intel Core i5/i7 or equivalent
Memory	RAM for processing datasets	Minimum 8 GB RAM
Storage	Space for datasets, trained model, and project files	256 GB SSD or higher
Software		
Frameworks	Backend Framework	Flask
Libraries	Machine Learning & Data Processing	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Visual Studio Code / Jupyter Notebook
Data		
Data	Source, size, format	Well-Being Dataset (CSV format), containing health, lifestyle, and demographic features used for training and testing the Machine Learning model.

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